

Mathematical Foundations of Machine Learning

Assignment 5 Solutions

Question 1 (20 points)

Work in $\mathbb{R}^4$ with the standard inner product. Let

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 0 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 1 \\ 1 \\ 0 \\ 1 \end{bmatrix}, \quad \mathbf{v}_3 = \begin{bmatrix} 1 \\ 0 \\ 1 \\ 1 \end{bmatrix}, \quad \mathbf{w} = \begin{bmatrix} 2 \\ 1 \\ 0 \\ 1 \end{bmatrix}.$$

1. **(10 points)** Apply Gram-Schmidt to $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ to obtain an orthonormal basis $\{\mathbf{q}_1, \mathbf{q}_2, \mathbf{q}_3\}$ for $U = \text{span}\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$.
2. **(6 points)** Using your orthonormal basis, compute the least-squares approximation $\hat{\mathbf{w}} \in U$ of $\mathbf{w}$, i.e.,

$$\hat{\mathbf{w}} = \arg \min_{\mathbf{u} \in U} \|\mathbf{w} - \mathbf{u}\|_2.$$

3. **(4 points)** Let $\mathbf{r} = \mathbf{w} - \hat{\mathbf{w}}$. Compute $\mathbf{r}$ and show that it is orthogonal to U (it suffices to check $\langle \mathbf{r}, \mathbf{q}_i \rangle = 0$ for $i = 1, 2, 3$). Also compute $\|\mathbf{r}\|_2^2$.

Solution:

1. Apply Gram-Schmidt to $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$.

Set $\mathbf{u}_1 = \mathbf{v}_1$, so $\|\mathbf{u}_1\|_2 = \sqrt{3}$ and

$$\mathbf{q}_1 = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 0 \end{bmatrix}.$$

Next,

$$\langle \mathbf{v}_2, \mathbf{q}_1 \rangle = \frac{1}{\sqrt{3}}(1+1+0+0) = \frac{2}{\sqrt{3}}, \quad \mathbf{u}_2 = \mathbf{v}_2 - \langle \mathbf{v}_2, \mathbf{q}_1 \rangle \mathbf{q}_1 = \mathbf{v}_2 - \frac{2}{3} \mathbf{v}_1 = \begin{bmatrix} \frac{1}{3} \\ \frac{1}{3} \\ \frac{2}{3} \\ 1 \end{bmatrix}.$$

Then

$$\|\mathbf{u}_2\|_2^2 = \frac{1}{9} + \frac{1}{9} + \frac{4}{9} + 1 = \frac{5}{3}, \quad \|\mathbf{u}_2\|_2 = \sqrt{\frac{5}{3}},$$

so

$$\mathbf{q}_2 = \frac{\mathbf{u}_2}{\|\mathbf{u}_2\|_2} = \frac{1}{\sqrt{15}} \begin{bmatrix} 1 \\ 1 \\ -2 \\ 3 \end{bmatrix}.$$

For $\mathbf{v}_3$,

$$\langle \mathbf{v}_3, \mathbf{q}_1 \rangle = \frac{1}{\sqrt{3}}(1+0+1+0) = \frac{2}{\sqrt{3}}, \quad \langle \mathbf{v}_3, \mathbf{q}_2 \rangle = \frac{1}{\sqrt{15}}(1+0-2+3) = \frac{2}{\sqrt{15}},$$

so

$$\mathbf{u}_3 = \mathbf{v}_3 - \langle \mathbf{v}_3, \mathbf{q}_1 \rangle \mathbf{q}_1 - \langle \mathbf{v}_3, \mathbf{q}_2 \rangle \mathbf{q}_2 = \begin{bmatrix} \frac{1}{5} \\ \frac{4}{5} \\ \frac{3}{5} \\ \frac{3}{5} \end{bmatrix}.$$

Thus

$$\|\mathbf{u}_3\|_2^2 = \frac{1}{25} + \frac{16}{25} + \frac{9}{25} + \frac{9}{25} = \frac{7}{5}, \quad \|\mathbf{u}_3\|_2 = \sqrt{\frac{7}{5}},$$

and hence

$$\mathbf{q}_3 = \frac{\mathbf{u}_3}{\|\mathbf{u}_3\|_2} = \frac{1}{\sqrt{35}} \begin{bmatrix} 1 \\ -4 \\ 3 \\ 3 \end{bmatrix}.$$

Therefore, an orthonormal basis for

$$U = \text{span}\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$$

is

$$\mathbf{q}_1 = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 0 \end{bmatrix}, \quad \mathbf{q}_2 = \frac{1}{\sqrt{15}} \begin{bmatrix} 1 \\ 1 \\ -2 \\ 3 \end{bmatrix}, \quad \mathbf{q}_3 = \frac{1}{\sqrt{35}} \begin{bmatrix} 1 \\ -4 \\ 3 \\ 3 \end{bmatrix}.$$

2. Using the orthonormal basis $\{\mathbf{q}_1, \mathbf{q}_2, \mathbf{q}_3\}$, the least-squares approximation of $\mathbf{w}$ in U is

$$\hat{\mathbf{w}} = \sum_{i=1}^3 \langle \mathbf{w}, \mathbf{q}_i \rangle \mathbf{q}_i.$$

Compute the coefficients:

$$\langle \mathbf{w}, \mathbf{q}_1 \rangle = \frac{1}{\sqrt{3}}(2+1+0+0) = \sqrt{3},$$

$$\langle \mathbf{w}, \mathbf{q}_2 \rangle = \frac{1}{\sqrt{15}}(2+1+0+3) = \frac{6}{\sqrt{15}},$$

$$\langle \mathbf{w}, \mathbf{q}_3 \rangle = \frac{1}{\sqrt{35}}(2 - 4 + 0 + 3) = \frac{1}{\sqrt{35}}.$$

Hence

$$\hat{\mathbf{w}} = \sqrt{3} \mathbf{q}_1 + \frac{6}{\sqrt{15}} \mathbf{q}_2 + \frac{1}{\sqrt{35}} \mathbf{q}_3 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 0 \end{bmatrix} + \begin{bmatrix} \frac{2}{5} \\ \frac{2}{5} \\ \frac{4}{5} \\ -\frac{6}{5} \end{bmatrix} + \begin{bmatrix} \frac{1}{35} \\ -\frac{4}{35} \\ \frac{3}{35} \\ \frac{3}{35} \end{bmatrix}.$$

Therefore

$$\hat{\mathbf{w}} = \begin{bmatrix} \frac{10}{7} \\ \frac{9}{7} \\ \frac{2}{7} \\ \frac{9}{7} \end{bmatrix}.$$

3. The residual is

$$\mathbf{r} = \mathbf{w} - \hat{\mathbf{w}} = \begin{bmatrix} 2 \\ 1 \\ 0 \\ 1 \end{bmatrix} - \begin{bmatrix} \frac{10}{7} \\ \frac{9}{7} \\ \frac{2}{7} \\ \frac{9}{7} \end{bmatrix} = \begin{bmatrix} \frac{4}{7} \\ -\frac{2}{7} \\ -\frac{2}{7} \\ -\frac{2}{7} \end{bmatrix}.$$

Now check that $\mathbf{r}$ is orthogonal to each basis vector of U :

$$\langle \mathbf{r}, \mathbf{q}_1 \rangle = \frac{1}{\sqrt{3}} \left(\frac{4}{7} - \frac{2}{7} - \frac{2}{7} \right) = 0,$$

$$\langle \mathbf{r}, \mathbf{q}_2 \rangle = \frac{1}{\sqrt{15}} \left(\frac{4}{7} - \frac{2}{7} + 2 \cdot \frac{2}{7} - 3 \cdot \frac{2}{7} \right) = 0,$$

$$\langle \mathbf{r}, \mathbf{q}_3 \rangle = \frac{1}{\sqrt{35}} \left(\frac{4}{7} + 4 \cdot \frac{2}{7} - 3 \cdot \frac{2}{7} - 3 \cdot \frac{2}{7} \right) = 0.$$

Thus $\mathbf{r} \perp U$.

Finally,

$$\|\mathbf{r}\|_2^2 = \left(\frac{4}{7} \right)^2 + 3 \left(\frac{2}{7} \right)^2 = \frac{16}{49} + \frac{12}{49} = \frac{28}{49} = \frac{4}{7}.$$

Question 2 (20 points)

Let $\alpha \in \mathbb{R}$ and define

$$\mathbf{A}(\alpha) = \begin{bmatrix} \alpha & 1 & 1 \\ 1 & \alpha & 1 \\ 1 & 1 & \alpha \end{bmatrix}.$$

1. **(12 points)** Compute $\det(\mathbf{A}(\alpha))$ as a polynomial in α .
2. **(4 points)** For which values of α is $\mathbf{A}(\alpha)$ invertible?

3. **(4 points)** For each value of α where $\mathbf{A}(\alpha)$ is singular, describe $\ker(\mathbf{A}(\alpha))$ (give its dimension and a basis).

Solution:

1. Apply $R_2 \leftarrow R_2 - R_1$ and $R_3 \leftarrow R_3 - R_1$ (determinant unchanged):

$$\det(\mathbf{A}(\alpha)) = \det \begin{bmatrix} \alpha & 1 & 1 \\ 1 - \alpha & \alpha - 1 & 0 \\ 1 - \alpha & 0 & \alpha - 1 \end{bmatrix}.$$

Factor $(\alpha - 1)$ out of rows 2 and 3 (note $1 - \alpha = -(\alpha - 1)$):

$$\det(\mathbf{A}(\alpha)) = (\alpha - 1)^2 \det \begin{bmatrix} \alpha & 1 & 1 \\ -1 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix}.$$

Now expand the remaining determinant along the first row:

$$\begin{aligned} \det \begin{bmatrix} \alpha & 1 & 1 \\ -1 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix} &= \alpha \det \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - 1 \det \begin{bmatrix} -1 & 0 \\ -1 & 1 \end{bmatrix} + 1 \det \begin{bmatrix} -1 & 1 \\ -1 & 0 \end{bmatrix} \\ &= \alpha(1) - 1(-1) + 1(1) = \alpha + 2. \end{aligned}$$

Hence

$$\det(\mathbf{A}(\alpha)) = (\alpha - 1)^2(\alpha + 2).$$

2. $\mathbf{A}(\alpha)$ is invertible iff $\det(\mathbf{A}(\alpha)) \neq 0$, i.e.

$$(\alpha - 1)^2(\alpha + 2) \neq 0 \iff \alpha \neq 1 \text{ and } \alpha \neq -2.$$

3. If $\alpha = 1$, then $\mathbf{A}(1)$ is the all-ones matrix. For any $\mathbf{x} \in \mathbb{R}^3$,

$$\mathbf{A}(1)\mathbf{x} = (x_1 + x_2 + x_3) \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}.$$

Thus $\mathbf{A}(1)\mathbf{x} = \mathbf{0}$ iff $x_1 + x_2 + x_3 = 0$. Therefore

$$\ker(\mathbf{A}(1)) = \{\mathbf{x} \in \mathbb{R}^3 : x_1 + x_2 + x_3 = 0\},$$

which has dimension 2, with for example basis

$$\begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}, \quad \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}.$$

If $\alpha = -2$, then

$$\mathbf{A}(-2) \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} -2 + 1 + 1 \\ 1 - 2 + 1 \\ 1 + 1 - 2 \end{bmatrix} = \mathbf{0},$$

so $\mathbf{1} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \in \ker(\mathbf{A}(-2))$. Since $\alpha = -2$ is a simple root of $\det(\mathbf{A}(\alpha))$, the nullity is 1, hence

$$\ker(\mathbf{A}(-2)) = \text{span} \left\{ \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \right\}.$$

Question 3 (12 points)

Let $\mathbf{A} = (a_{ij})$ be an $n \times n$ matrix. Prove that if the sum of the absolute values of the elements in each row of $\mathbf{A}$ is less than 1, then the magnitude of every eigenvalue of $\mathbf{A}$ is less than 1.

Solution:

Let λ be an arbitrary eigenvalue of the matrix $\mathbf{A}$, and let $\mathbf{x} = (x_1, x_2, \dots, x_n)^T$ be its corresponding non-zero eigenvector. Thus, we have the characteristic equation:

$$\mathbf{A}\mathbf{x} = \lambda\mathbf{x}$$

Consider the component of $\mathbf{x}$ with the largest absolute value. Let this be the k -th component, denoted by x_k , such that:

$$|x_k| = \max_{1 \leq i \leq n} |x_i|$$

Since $\mathbf{x} \neq \mathbf{0}$, it follows that $|x_k| > 0$.

From the k -th row of the matrix equation $\mathbf{A}\mathbf{x} = \lambda\mathbf{x}$, we obtain:

$$\sum_{j=1}^n a_{kj}x_j = \lambda x_k$$

Taking the absolute value of both sides and applying the triangle inequality:

$$|\lambda||x_k| = \left| \sum_{j=1}^n a_{kj}x_j \right| \leq \sum_{j=1}^n |a_{kj}||x_j|$$

Since $|x_k|$ is the maximum magnitude among all components, we know that $|x_j| \leq |x_k|$ for all $j = 1, \dots, n$. We can substitute this into the inequality:

$$|\lambda||x_k| \leq \left(\sum_{j=1}^n |a_{kj}| \right) |x_k|$$

Dividing both sides by $|x_k|$ (which is strictly positive):

$$|\lambda| \leq \sum_{j=1}^n |a_{kj}|$$

According to the given condition, the sum of the absolute values of the elements in each row is less than 1. Specifically for the k -th row:

$$\sum_{j=1}^n |a_{kj}| < 1$$

Therefore, we conclude that the magnitude of the eigenvalue must satisfy:

$$|\lambda| < 1$$

Question 4 (18 points)

Let $a_i \in \mathbb{R}$ ($i = 1, 2, \dots, n$) satisfy $\sum_{i=1}^n a_i = 0$. Define matrix $\mathbf{A}$ by its entries $A_{ij} = a_i a_j + 1$.

1. **(3 points)** Prove that $\mathbf{A} = \mathbf{a}\mathbf{a}^T + \mathbf{1}\mathbf{1}^T$, where $\mathbf{a} = [a_1, \dots, a_n]^T$ and $\mathbf{1} = [1, \dots, 1]^T$.
2. **(15 points)** Using the fact that $\mathbf{a}$ and $\mathbf{1}$ are orthogonal, find all eigenvalues of $\mathbf{A}$ and their respective algebraic multiplicities.

Solution:

1. The (i, j) -entry of $\mathbf{a}\mathbf{a}^T$ is $a_i a_j$, and the (i, j) -entry of $\mathbf{1}\mathbf{1}^T$ is 1, where

$$\mathbf{a} = \begin{bmatrix} a_1 \\ \vdots \\ a_n \end{bmatrix}, \quad \mathbf{1} = \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix}.$$

Therefore the (i, j) -entry of $\mathbf{a}\mathbf{a}^T + \mathbf{1}\mathbf{1}^T$ is

$$a_i a_j + 1 = A_{ij}.$$

Hence

$$\mathbf{A} = \mathbf{a}\mathbf{a}^T + \mathbf{1}\mathbf{1}^T.$$

2. Since $\sum_{i=1}^n a_i = 0$, we have

$$\mathbf{1}^T \mathbf{a} = \sum_{i=1}^n a_i = 0,$$

so $\mathbf{1}$ and $\mathbf{a}$ are orthogonal.

Let

$$s := \mathbf{a}^T \mathbf{a} = \sum_{i=1}^n a_i^2.$$

For any $\mathbf{x} \in \mathbb{R}^n$,

$$\mathbf{A}\mathbf{x} = (\mathbf{a}\mathbf{a}^T + \mathbf{1}\mathbf{1}^T)\mathbf{x} = \mathbf{a}(\mathbf{a}^T \mathbf{x}) + \mathbf{1}(\mathbf{1}^T \mathbf{x}).$$

First, apply $\mathbf{A}$ to $\mathbf{1}$:

$$\mathbf{A}\mathbf{1} = \mathbf{a}(\mathbf{a}^T \mathbf{1}) + \mathbf{1}(\mathbf{1}^T \mathbf{1}) = \mathbf{a} \cdot 0 + \mathbf{1} \cdot n = n\mathbf{1}.$$

So $\mathbf{1}$ is an eigenvector with eigenvalue n .

Next, apply $\mathbf{A}$ to $\mathbf{a}$:

$$\mathbf{A}\mathbf{a} = \mathbf{a}(\mathbf{a}^T \mathbf{a}) + \mathbf{1}(\mathbf{1}^T \mathbf{a}) = s\mathbf{a} + 0 = s\mathbf{a}.$$

Thus, if $\mathbf{a} \neq \mathbf{0}$, then $\mathbf{a}$ is an eigenvector with eigenvalue $s = \sum_{i=1}^n a_i^2$.

Now consider the subspace

$$U := \{\mathbf{x} \in \mathbb{R}^n : \mathbf{1}^T \mathbf{x} = 0, \mathbf{a}^T \mathbf{x} = 0\}.$$

For every $\mathbf{x} \in U$,

$$\mathbf{A}\mathbf{x} = \mathbf{a}(\mathbf{a}^T \mathbf{x}) + \mathbf{1}(\mathbf{1}^T \mathbf{x}) = \mathbf{0},$$

so every vector in U is an eigenvector with eigenvalue 0.

We now distinguish cases.

Case 1: $\mathbf{a} = \mathbf{0}$.

Then

$$\mathbf{A} = \mathbf{1}\mathbf{1}^T.$$

In this case,

$$\mathbf{A}\mathbf{1} = n\mathbf{1},$$

and every vector orthogonal to $\mathbf{1}$ is an eigenvector with eigenvalue 0. Hence the eigenvalues are

n with algebraic multiplicity 1, 0 with algebraic multiplicity $n-1$.

Case 2: $\mathbf{a} \neq \mathbf{0}$ and $s \neq n$.

Since $\mathbf{1} \perp \mathbf{a}$, the vectors $\mathbf{1}$ and $\mathbf{a}$ are linearly independent. Also,

$$\dim(U) = n - 2.$$

Choose a basis $\{\mathbf{u}_1, \dots, \mathbf{u}_{n-2}\}$ of U . Then

$$\{\mathbf{1}, \mathbf{a}, \mathbf{u}_1, \dots, \mathbf{u}_{n-2}\}$$

is a basis of $\mathbb{R}^n$ consisting entirely of eigenvectors of $\mathbf{A}$, with eigenvalues

$$n, \quad s, \quad \underbrace{0, \dots, 0}_{n-2 \text{ times}}.$$

Therefore the eigenvalues are

$$\begin{aligned} n & \text{ with algebraic multiplicity } 1, \\ \sum_{i=1}^n a_i^2 & \text{ with algebraic multiplicity } 1, \\ 0 & \text{ with algebraic multiplicity } n-2. \end{aligned}$$

Case 3: $\mathbf{a} \neq \mathbf{0}$ and $s = n$.

In this case both $\mathbf{1}$ and $\mathbf{a}$ are eigenvectors for the same eigenvalue n . Since they are orthogonal and nonzero, they are linearly independent. Again, if $\{\mathbf{u}_1, \dots, \mathbf{u}_{n-2}\}$ is a basis of U , then

$$\{\mathbf{1}, \mathbf{a}, \mathbf{u}_1, \dots, \mathbf{u}_{n-2}\}$$

is a basis of eigenvectors, and the corresponding eigenvalues are

$$n, \quad n, \quad \underbrace{0, \dots, 0}_{n-2 \text{ times}}.$$

Therefore the eigenvalues are

$$n \quad \text{with algebraic multiplicity } 2, \quad 0 \quad \text{with algebraic multiplicity } n-2.$$

Question 5 (20 points)

Let

$$\mathbf{B} = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 4 \end{bmatrix}.$$

1. **(4 points)** Compute the characteristic polynomial $p_{\mathbf{B}}(\lambda)$ and determine the eigenvalues of $\mathbf{B}$ with their algebraic multiplicities.
2. **(10 points)** For each eigenvalue λ , find a basis for the eigenspace E_{λ} .
3. **(6 points)** Determine the geometric multiplicity of each eigenvalue. Does the set of all eigenvectors of $\mathbf{B}$ span $\mathbb{R}^3$? If not, exhibit a vector in $\mathbb{R}^3$ that is not in the span of the eigenvectors.

Solution:

1. $\mathbf{B} - \lambda \mathbf{I}_3$ is upper triangular:

$$\mathbf{B} - \lambda \mathbf{I}_3 = \begin{bmatrix} 1 - \lambda & 1 & 0 \\ 0 & 1 - \lambda & 1 \\ 0 & 0 & 4 - \lambda \end{bmatrix}.$$

Using the determinant formula for triangular matrices,

$$p_{\mathbf{B}}(\lambda) = \det(\mathbf{B} - \lambda \mathbf{I}_3) = (1 - \lambda)^2(4 - \lambda).$$

Therefore the eigenvalues are $\lambda = 1$ (algebraic multiplicity 2) and $\lambda = 4$ (algebraic multiplicity 1).

2. For $\lambda = 4$,

$$\mathbf{B} - 4\mathbf{I}_3 = \begin{bmatrix} -3 & 1 & 0 \\ 0 & -3 & 1 \\ 0 & 0 & 0 \end{bmatrix}.$$

The equations are $-3x_1 + x_2 = 0$ and $-3x_2 + x_3 = 0$, so $x_2 = 3x_1$ and $x_3 = 9x_1$. Thus

$$E_4 = \text{span} \left\{ \begin{bmatrix} 1 \\ 3 \\ 9 \end{bmatrix} \right\}.$$

For $\lambda = 1$,

$$\mathbf{B} - \mathbf{I}_3 = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 3 \end{bmatrix}.$$

The equations imply $x_2 = 0$ and $x_3 = 0$, while x_1 is free. Hence

$$E_1 = \text{span} \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \right\}.$$

3. We have $\dim(E_4) = 1$ and $\dim(E_1) = 1$, so the geometric multiplicity of $\lambda = 4$ is 1 and the geometric multiplicity of $\lambda = 1$ is 1. Thus the span of all eigenvectors has dimension at most 2, so it cannot equal $\mathbb{R}^3$.

Concretely, the eigenvectors span $\text{span}\{\mathbf{e}_1, \mathbf{v}\}$ where $\mathbf{e}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$ and $\mathbf{v} =$

$\begin{bmatrix} 1 \\ 3 \\ 9 \end{bmatrix}$. Take $\mathbf{e}_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$. If $\mathbf{e}_2 = a\mathbf{e}_1 + b\mathbf{v}$ then

$$\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} = a \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + b \begin{bmatrix} 1 \\ 3 \\ 9 \end{bmatrix} = \begin{bmatrix} a + b \\ 3b \\ 9b \end{bmatrix},$$

which would force $3b = 1$ and $9b = 0$, a contradiction. Hence $\mathbf{e}_2$ is not in the span of the eigenvectors.

Question 6 (10 points)

Let $\mathbf{A}$ be an $n \times n$ matrix. Prove that if every non-zero n -dimensional column vector is an eigenvector of $\mathbf{A}$, then $\mathbf{A}$ is a scalar matrix, i.e., $\mathbf{A} = k\mathbf{I}_n$ for some scalar k .

Solution:

Consider any two linearly independent vectors $\mathbf{v}_1$ and $\mathbf{v}_2$ in $\mathbb{R}^n$. According to the given condition, both $\mathbf{v}_1$, $\mathbf{v}_2$, and their sum $\mathbf{v}_1 + \mathbf{v}_2$ are eigenvectors of $\mathbf{A}$. Thus, there exist scalars λ_1, λ_2 , and λ_3 such that:

$$\mathbf{A}\mathbf{v}_1 = \lambda_1\mathbf{v}_1 \quad (1)$$

$$\mathbf{A}\mathbf{v}_2 = \lambda_2\mathbf{v}_2 \quad (2)$$

$$\mathbf{A}(\mathbf{v}_1 + \mathbf{v}_2) = \lambda_3(\mathbf{v}_1 + \mathbf{v}_2) \quad (3)$$

Using the linearity of matrix multiplication, we expand the left side of (3):

$$\mathbf{A}(\mathbf{v}_1 + \mathbf{v}_2) = \mathbf{A}\mathbf{v}_1 + \mathbf{A}\mathbf{v}_2 = \lambda_1\mathbf{v}_1 + \lambda_2\mathbf{v}_2$$

Comparing this with the right side of (3), we have:

$$\lambda_1\mathbf{v}_1 + \lambda_2\mathbf{v}_2 = \lambda_3\mathbf{v}_1 + \lambda_3\mathbf{v}_2$$

Rearranging the terms:

$$(\lambda_1 - \lambda_3)\mathbf{v}_1 + (\lambda_2 - \lambda_3)\mathbf{v}_2 = \mathbf{0}$$

Since $\mathbf{v}_1$ and $\mathbf{v}_2$ are linearly independent, the coefficients must be zero:

$$\lambda_1 - \lambda_3 = 0 \quad \text{and} \quad \lambda_2 - \lambda_3 = 0$$

This implies $\lambda_1 = \lambda_2 = \lambda_3 = k$.

Because this holds for any pair of linearly independent vectors, all eigenvalues associated with any vector in $\mathbb{R}^n$ must be the same constant k . Therefore, for every vector $\mathbf{v} \in \mathbb{R}^n$:

$$\mathbf{A}\mathbf{v} = k\mathbf{v} = (k\mathbf{I}_n)\mathbf{v}$$

It follows that $\mathbf{A} = k\mathbf{I}_n$, which means $\mathbf{A}$ is a scalar matrix.

Bonus (10 points)

Let $\mathbf{A} \in \mathbb{R}^{n \times n}$ be a symmetric matrix. Prove that eigenvectors corresponding to distinct eigenvalues of a symmetric matrix are linearly independent.

Solution:

1. Since $\mathbf{A}$ is symmetric,

$$(\mathbf{Ax})^\top \mathbf{y} = \mathbf{x}^\top \mathbf{A}^\top \mathbf{y} = \mathbf{x}^\top \mathbf{Ay}.$$

Using the eigenvector relations,

$$(\mathbf{Ax})^\top \mathbf{y} = (\lambda \mathbf{x})^\top \mathbf{y} = \lambda \mathbf{x}^\top \mathbf{y}, \quad \mathbf{x}^\top \mathbf{Ay} = \mathbf{x}^\top (\mu \mathbf{y}) = \mu \mathbf{x}^\top \mathbf{y}.$$

Therefore $\lambda \mathbf{x}^\top \mathbf{y} = \mu \mathbf{x}^\top \mathbf{y}$, i.e.

$$(\lambda - \mu) \mathbf{x}^\top \mathbf{y} = 0.$$

Since $\lambda \neq \mu$, we must have $\mathbf{x}^\top \mathbf{y} = 0$.

2. If $\mathbf{x}_1, \dots, \mathbf{x}_k$ are eigenvectors corresponding to *distinct* eigenvalues of a symmetric matrix, then by part (a) they are pairwise orthogonal. A set of nonzero pairwise orthogonal vectors is linearly independent, so $\{\mathbf{x}_1, \dots, \mathbf{x}_k\}$ is linearly independent.